

A vibrant rainbow arches over a green field with a red barn in the distance. The sky is a mix of blue and purple, suggesting a sunset or sunrise. The rainbow is the central focus, with its colors clearly visible. The field in the foreground is green and grassy, with some trees and a fence line visible. A red barn is visible on the horizon under the rainbow.

Om regnbågar

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Kulturnatten 19/9 2026

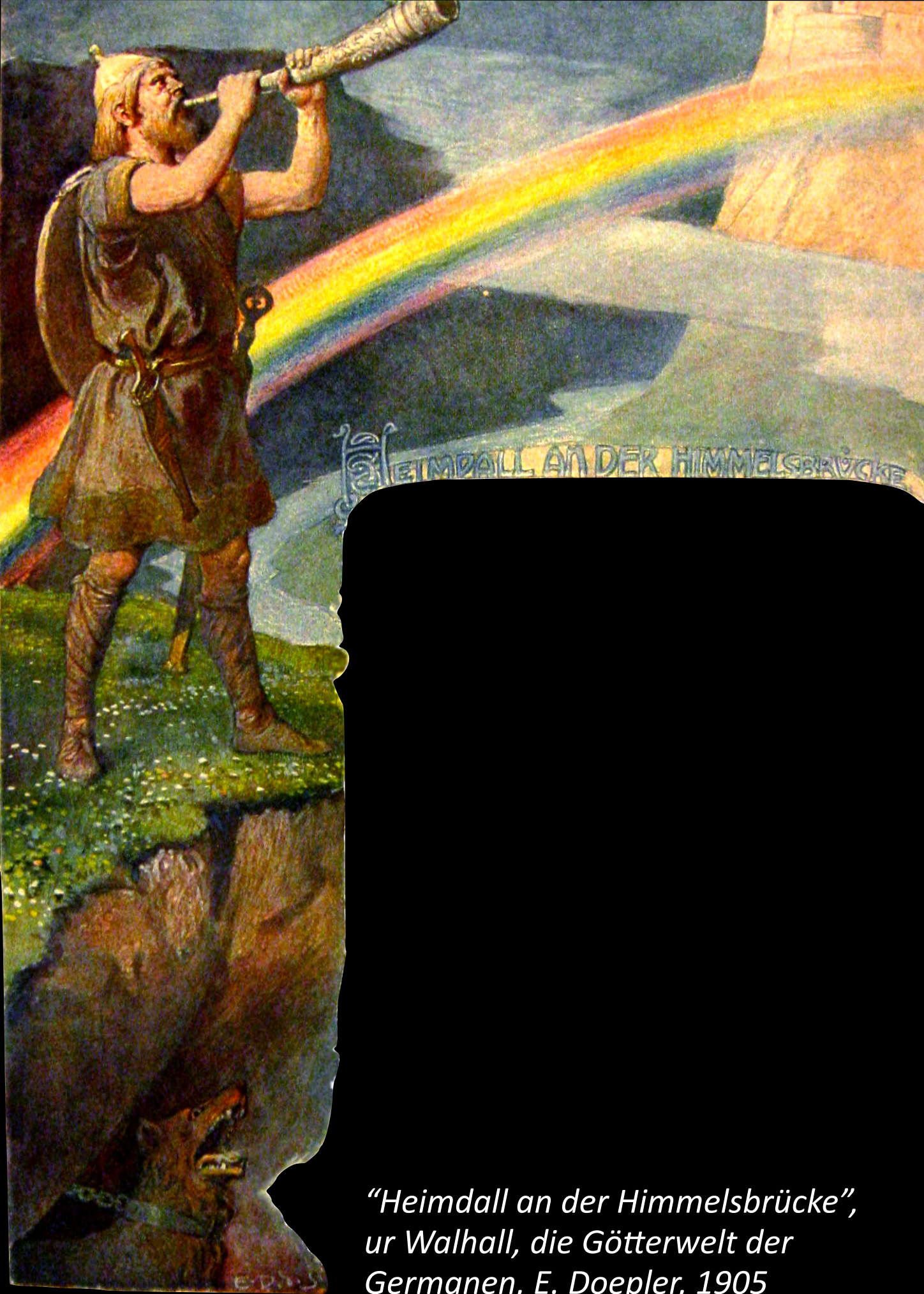


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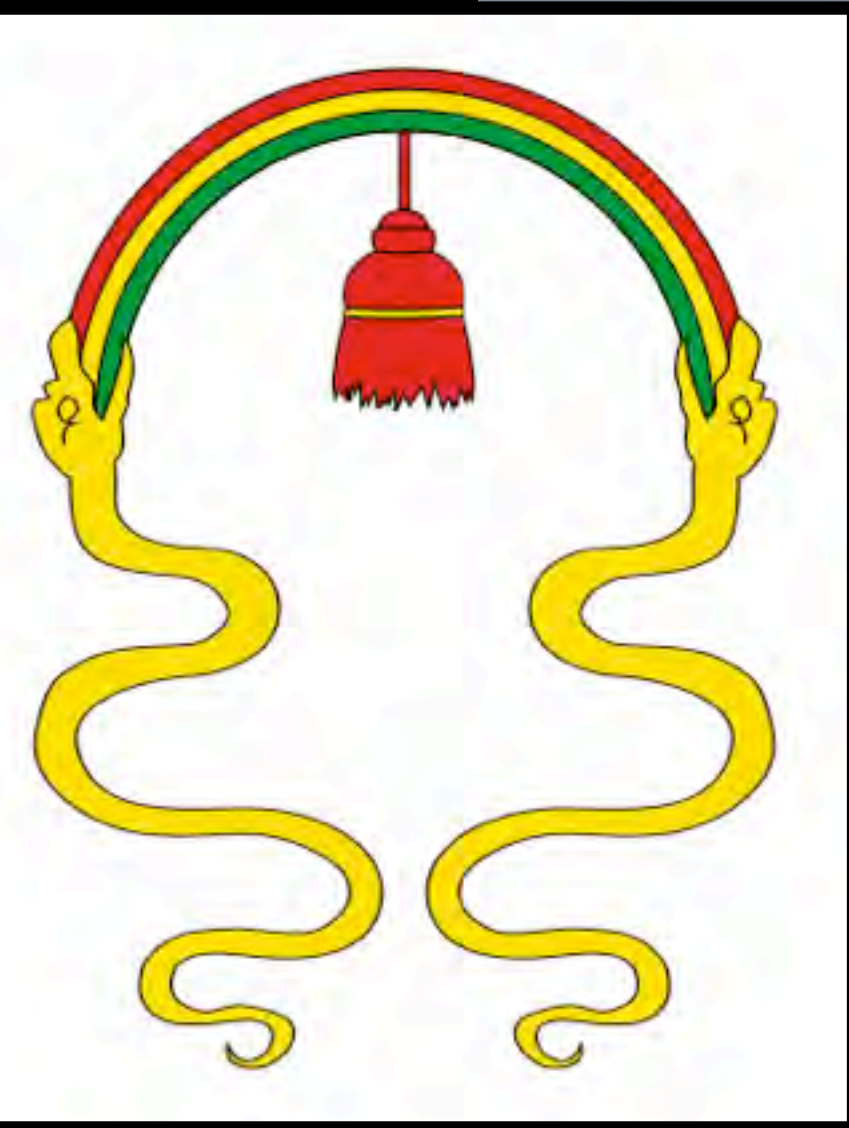


*"Heimdall an der Himmelsbrücke",
ur Walhall, die Götterwelt der
Germanen, E. Doepler, 1905*



*Historier om Noa – Regnbågen
St. Markus Basilika, Venedig, 1200-tal*

Cusco,
Peru





Petroglyf, Hopi-kulturen, Arizona, c. 700–1200

قوس قزح حادث اولدی مدوردوری تمام طاعنه بن اشاعی ایندیکه و این
کوجلدی وصاری اولدی سحابه که ایرشدم مضمحل اولدی والله اعلم **وهذه صورته**





"Das Zeichen des Bundes", Augsburger Wunderzeichenbuch, c. 1550



“Landschap met rustende en musicerende landlieden; in de lucht een regenboog”, Peter Paul Rubens, c. 1632–35



"Arundel Castle on the River Arun, with a Rainbow", William Turner, 1824–25



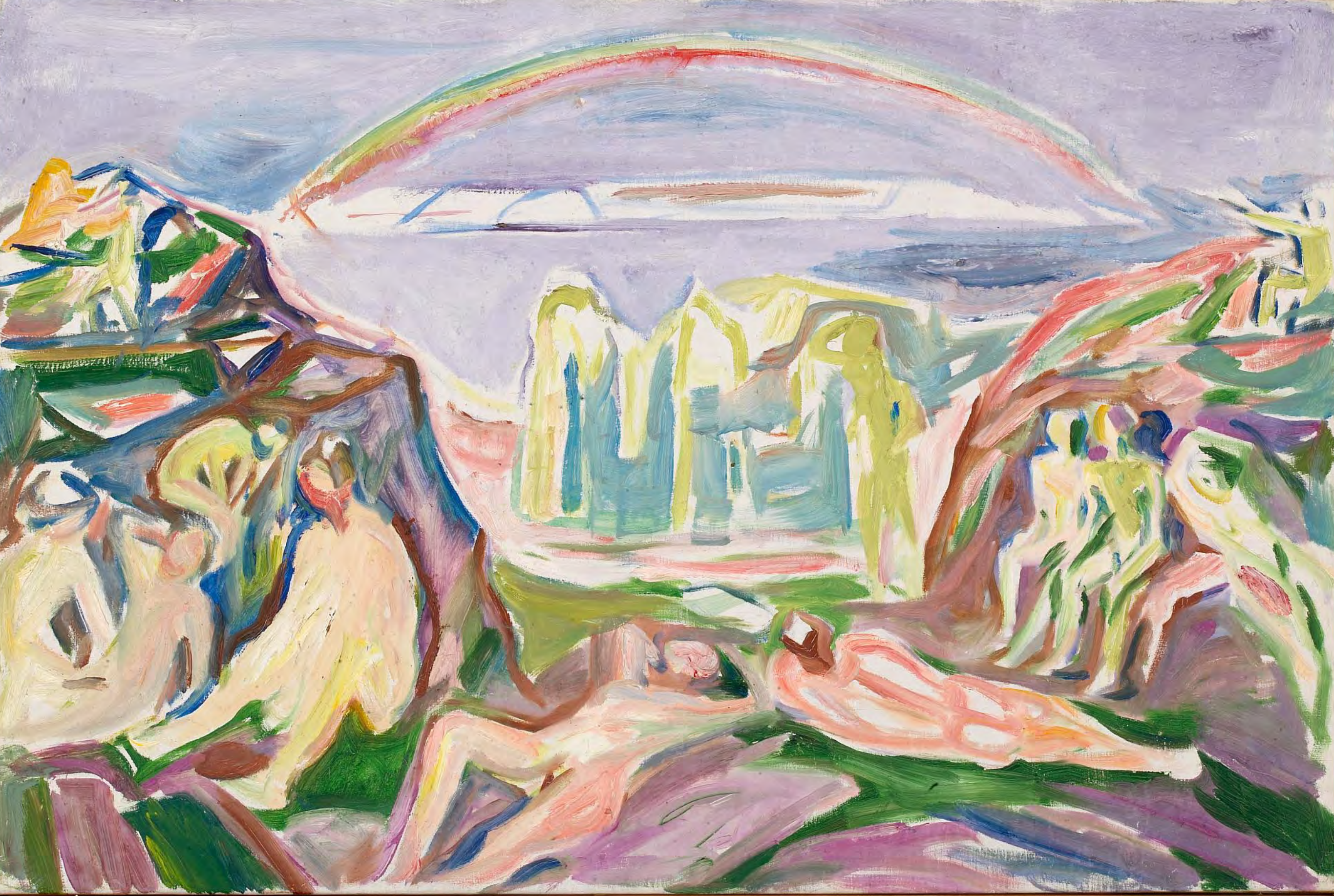
“Itáliai tájkép dupla szivárvánnyal (Italienskt landskap med dubbel regnbåge)”, Károly Markó, 1841



"Rainy Season in the Tropics", Frederic Edwin Church, 1866



"Regnbåge", Nils Kreuger, 1908



"Regnbuen", Edvard Munch, 1918–24



Rainbow (Ritchie Blackmore), Rising, 1976



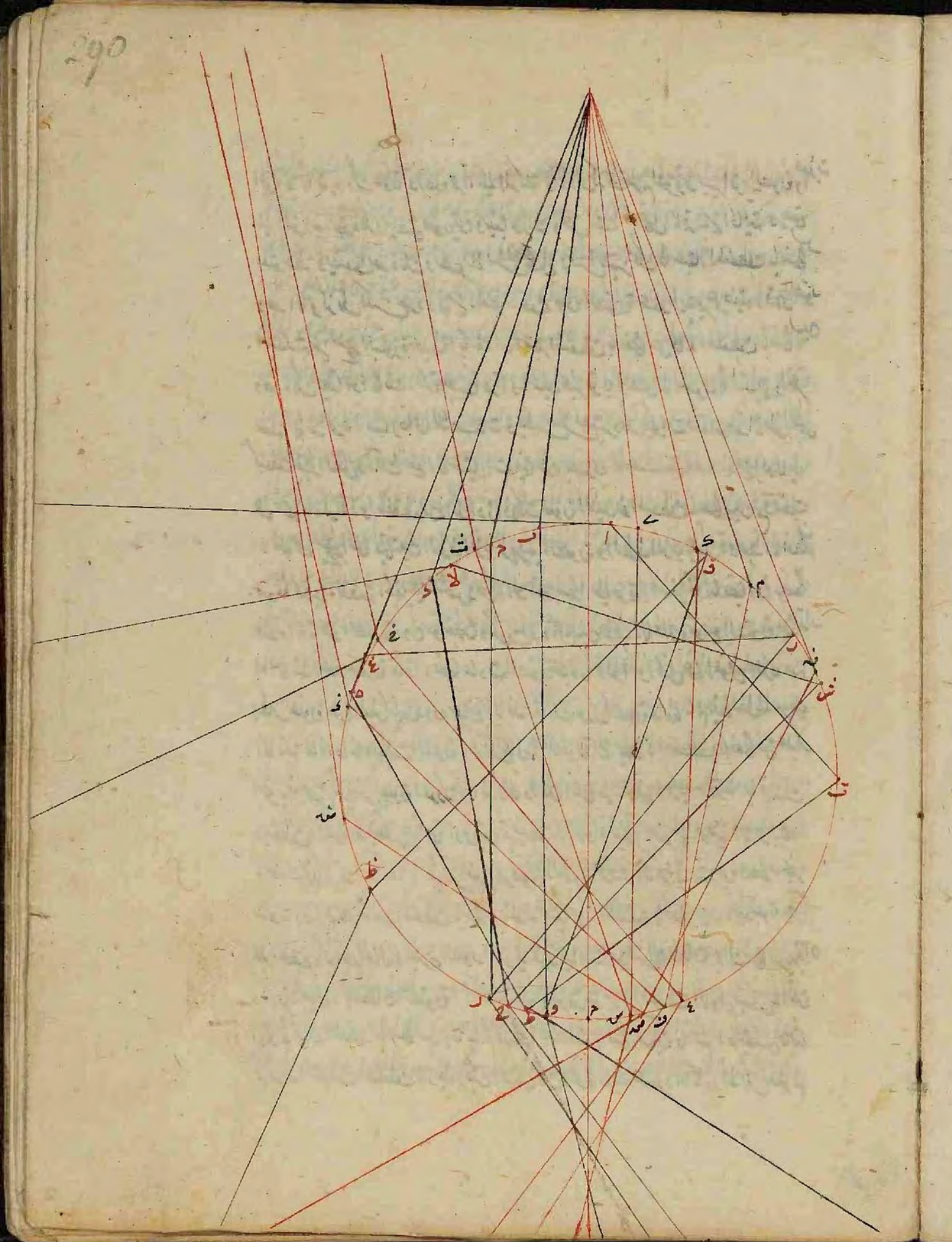
Pink Floyd, Dark Side of the Moon, 1973

Coola men uppenbart
fejkade bilder!

– hur ser vi det?



Demonstration



Veritasium (YouTube), 2024

Kamāl al-Dīn al-Fārisī: *Primär och sekundär regnbågskonstruktion*, ur *Revision av optiken enligt Ibn al-Haytham*, c. 1300



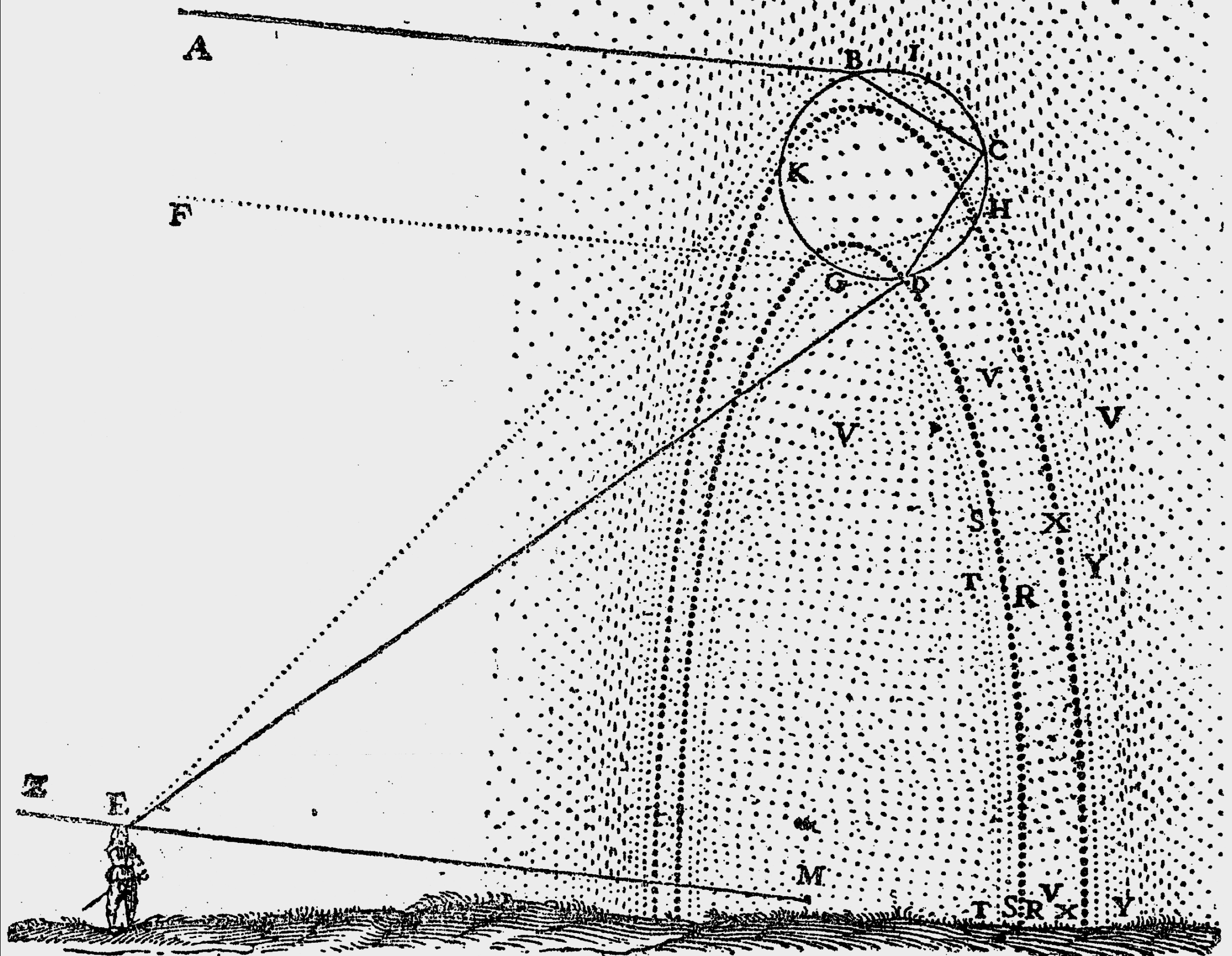


Fig. 13.

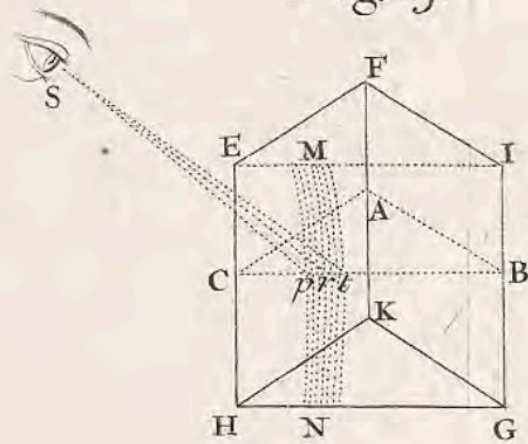


Fig. 14.

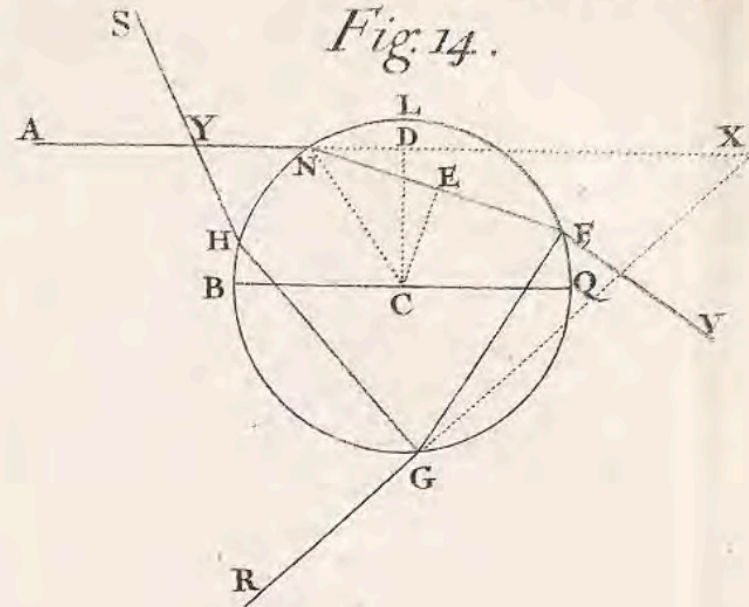


Fig. 15.

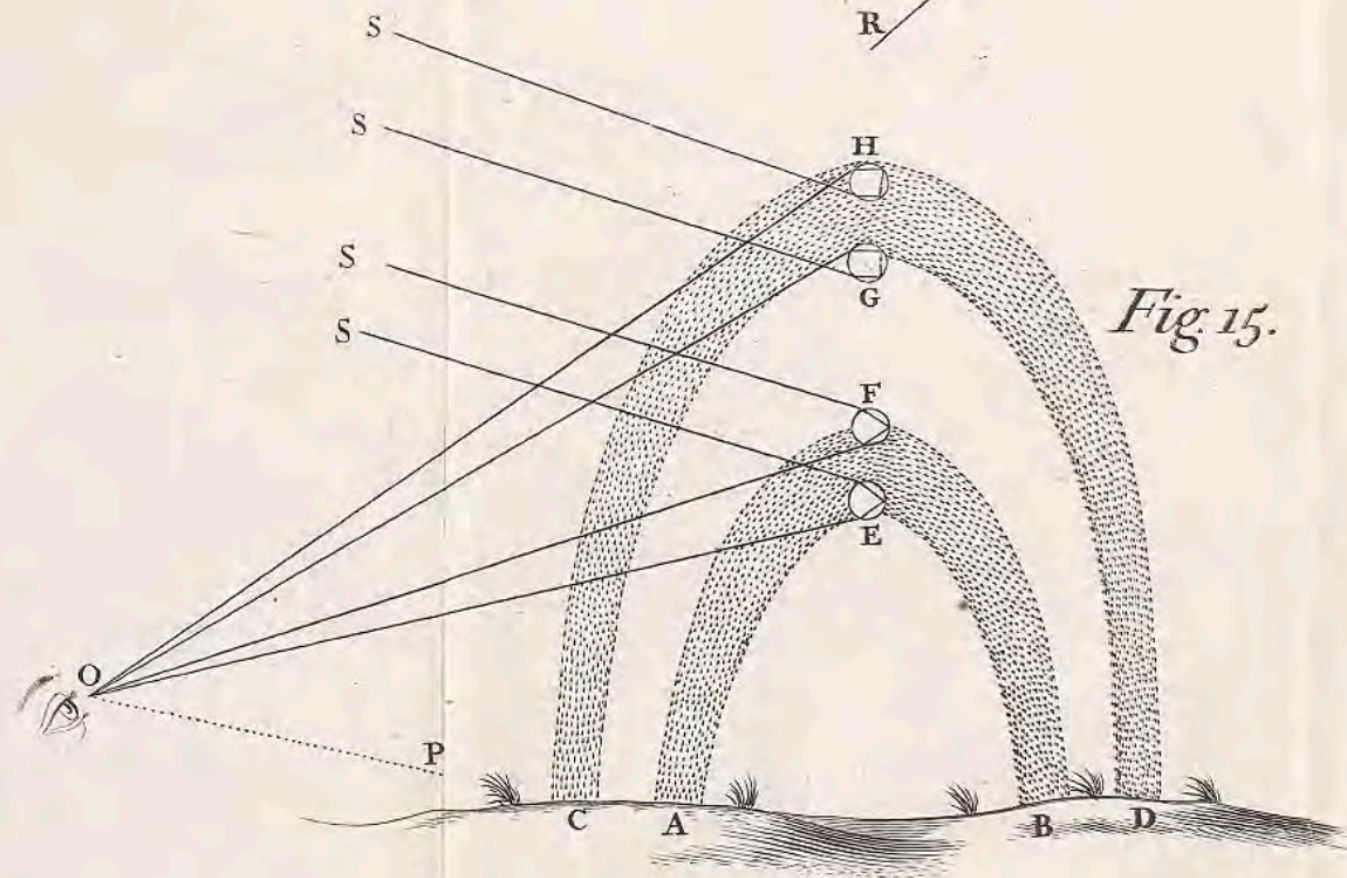
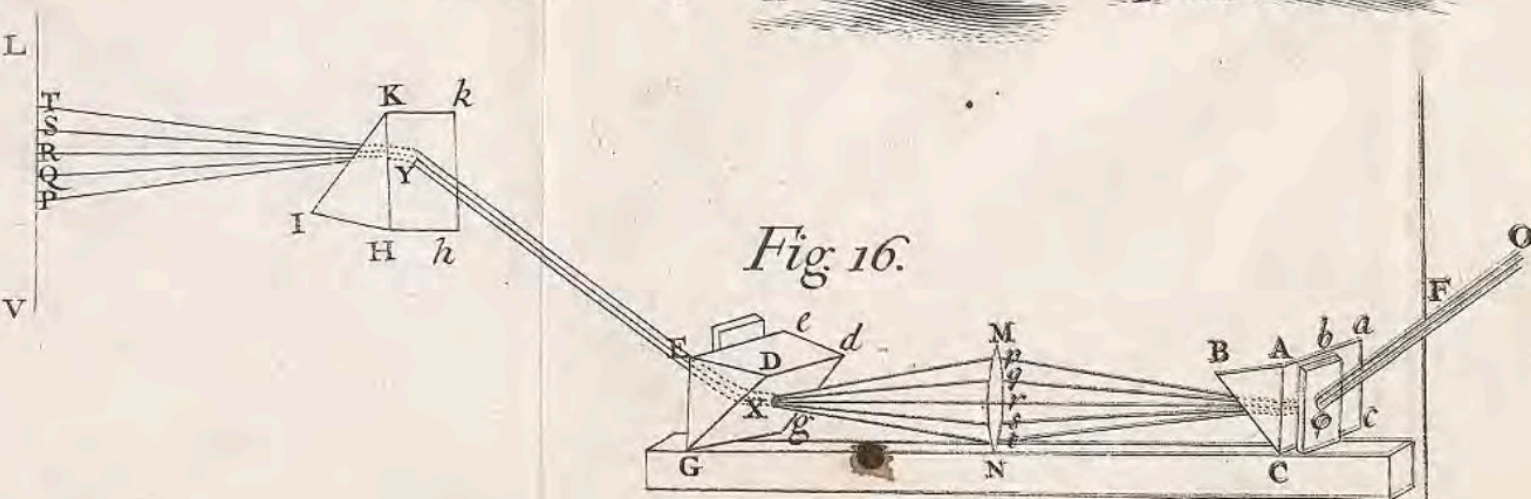


Fig. 16.

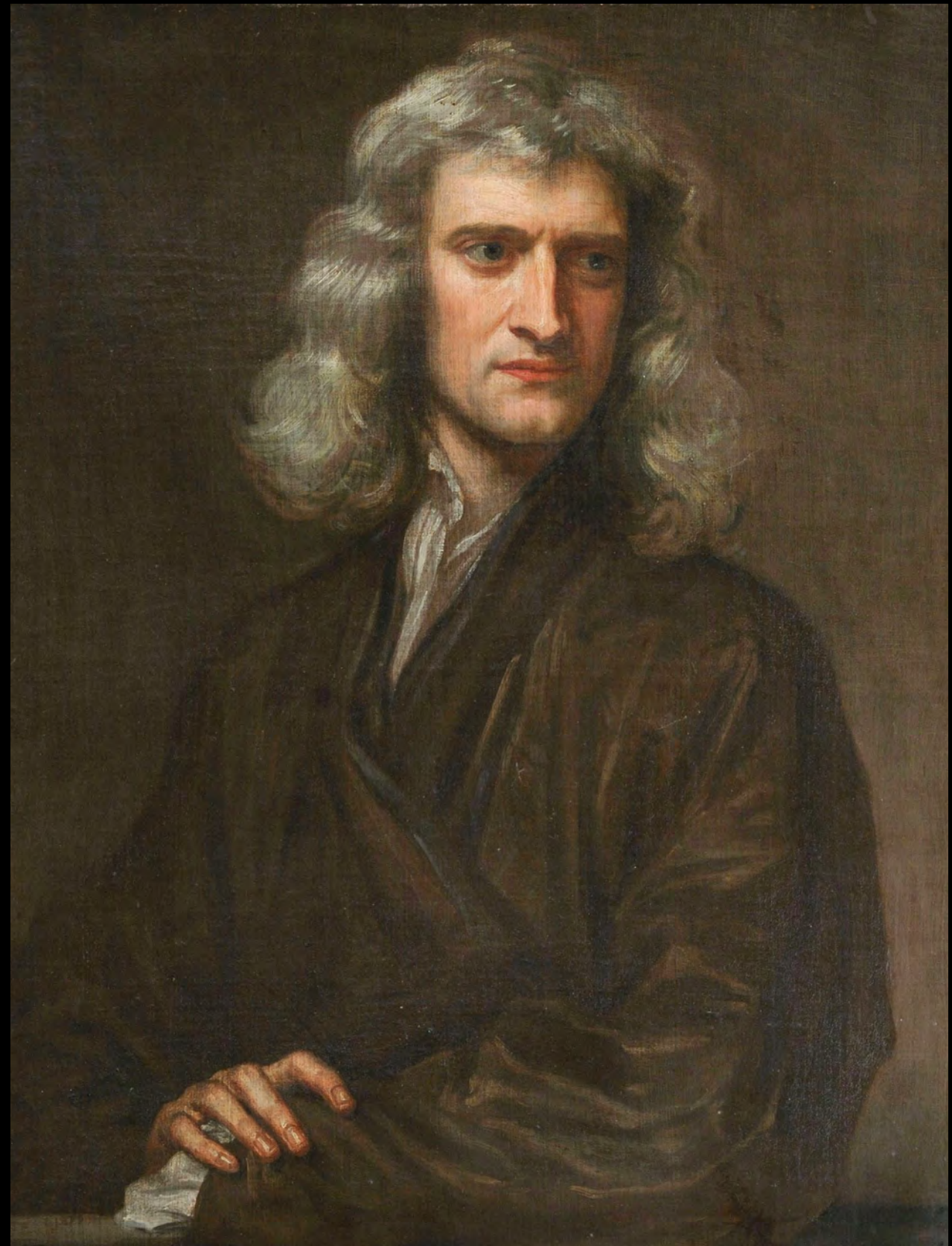


Isaac Newton
Opticks: Or, a Treatise of the Reflexions, Refractions, Inflexions and Colours of Light
 Samuel Smith and Benjamin Walford,
 London, 1704



Christiaan Huygens 1629 – 1695

“Ljus är en våg”



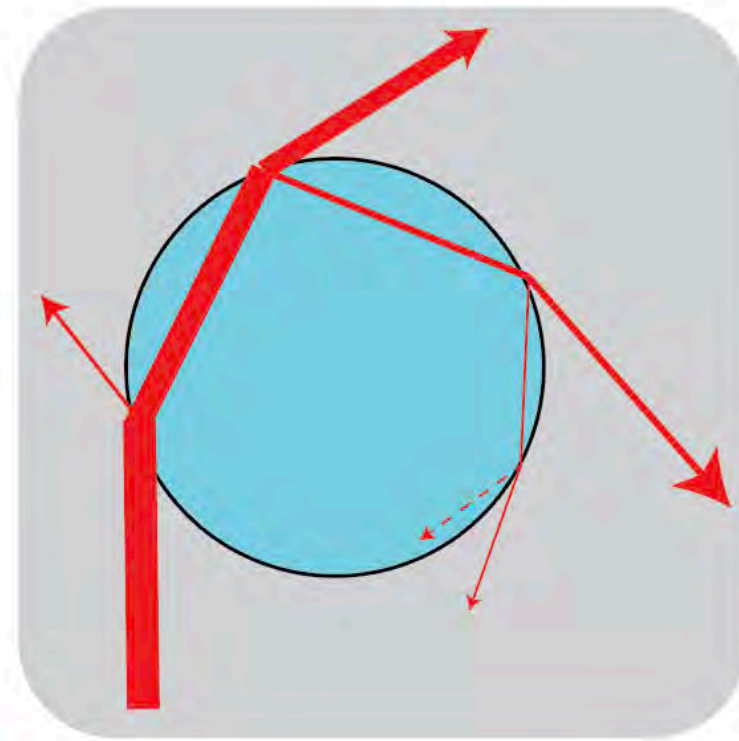
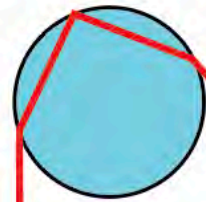
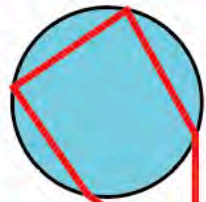
Isaac Newton 1643 – 1727

“Ljus är partiklar (*korpuskler*)”

Simulering 1–4

sekundär

primär

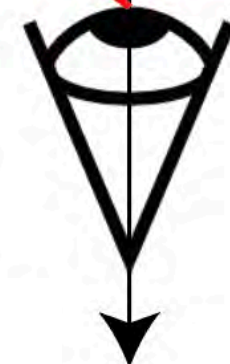


mot centrum
av regnbågen

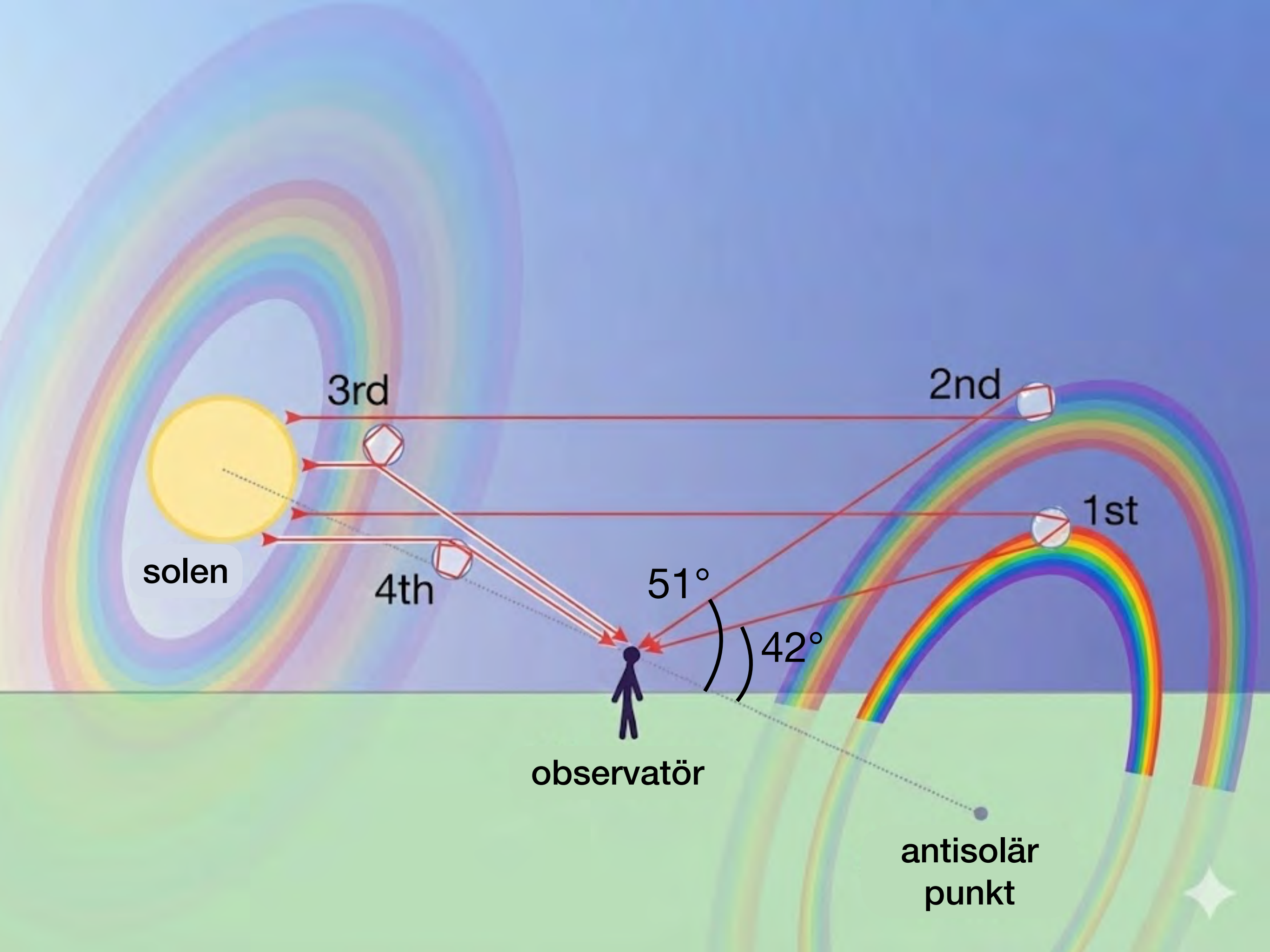
51°

42°

mot solen



Simulering 5–6



solen

3rd

2nd

1st

4th

51°

42°

observatör

antisolär
punkt



Alexanders mörka band:
himlens bakgrundsfärg!

Framför



MM 2023-09-01 08:14

Bakom



MM 2023-09-01 08:05



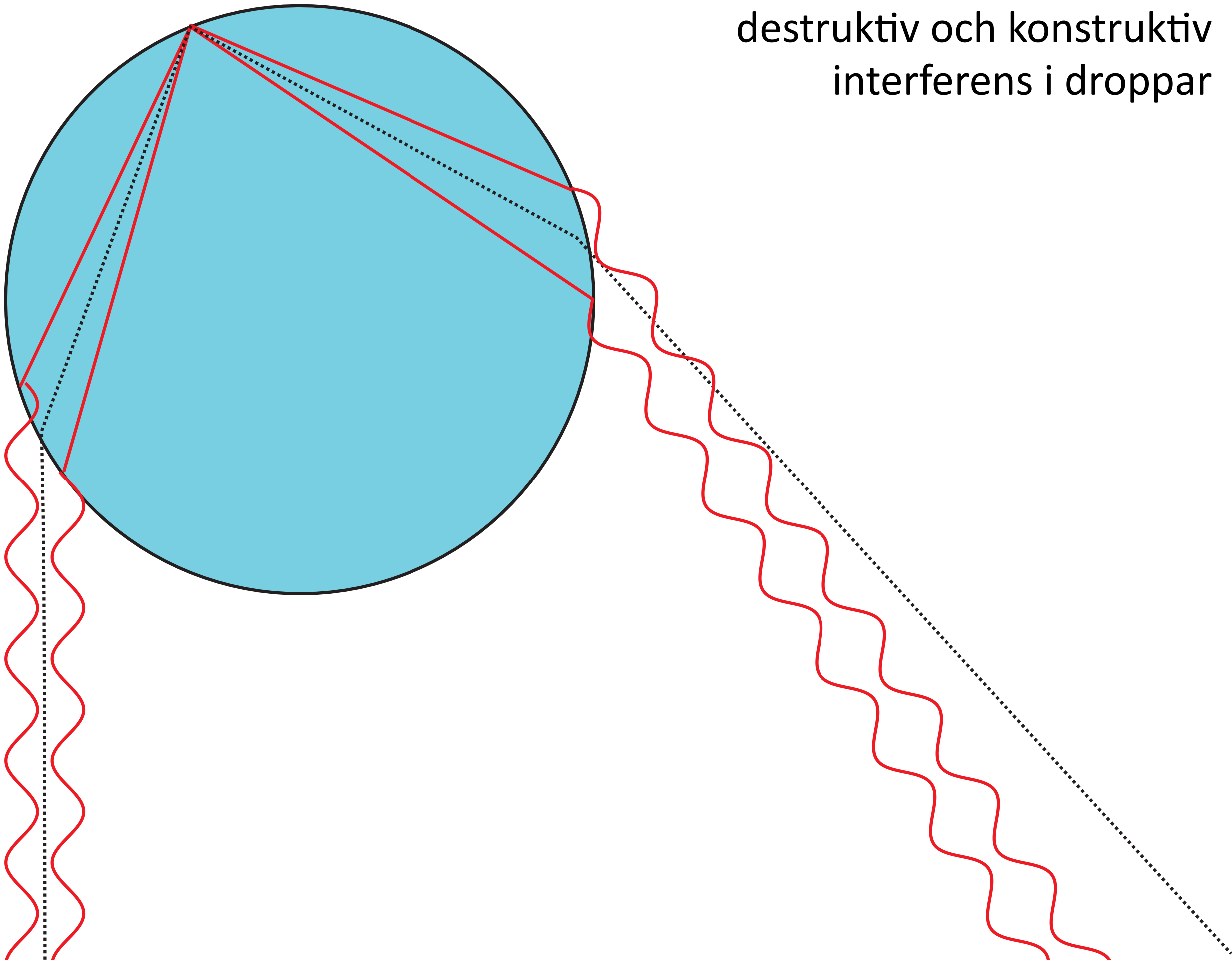
MM 2023-09-01 08:11

Supernumerär regnbåge

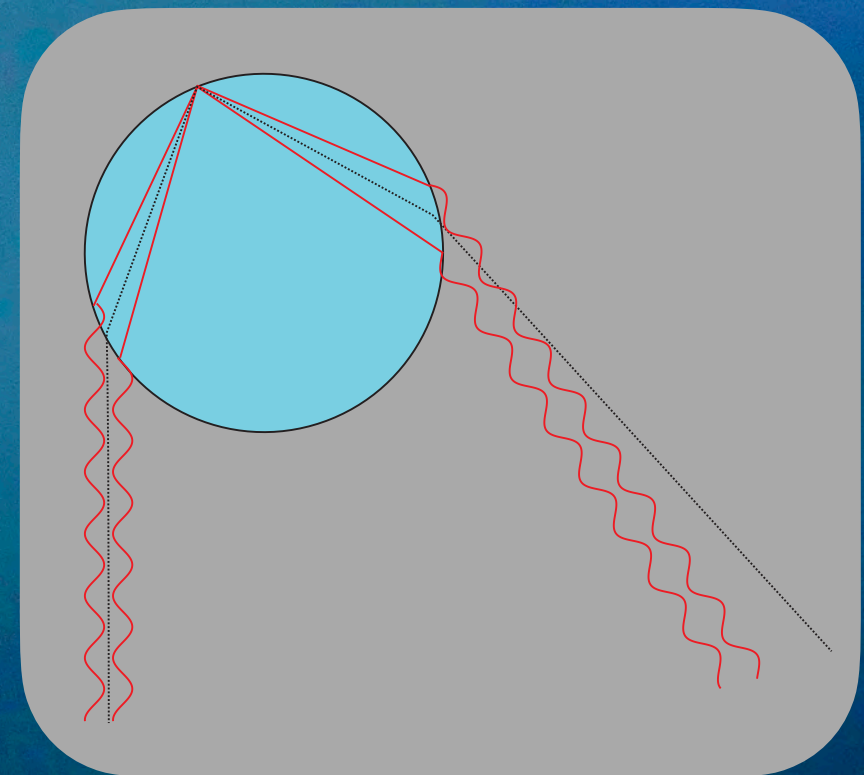


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Supernumerär regnbåge:
destruktiv och konstruktiv
interferens i droppar



Supernumerär regnbåge:
destruktiv och konstruktiv
interferens i droppar



$\approx 100 \mu\text{m}$ droppar
med variation $\pm 8 \%$

Dimbåge:
Färgerna blandas i
mycket små droppar

$\approx 20 \mu\text{m}$ droppar
med variation $\pm 30 \%$

MM 2015-09-27 08:01

Månbåge över Hawaii



Korona runt månen:
inte en regnbåge!
– Diffraktion från droppar



$\approx 8 \mu\text{m}$ droppar
med variation $\pm 10 \%$

Regnbåge efter
solnedgång



MM 2023-08-17 20:37



MM 2024-06-12 19:38